



İzmir Institute of Technology

CHE 222
Fluid Mechanics
Summer 2024-2025

TERM PROJECT
REPORT

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Date: 24.08.2025

İzmir, 2025

Properties and Calculations

$$\rho_{Biodiesel} = 883 \text{ kg/m}^3 [1]$$

$$\mu_{Biodiesel} = 0.0065 \text{ Pa.s} [1]$$

$$\text{fraction of Biodiesel} = 0.935$$

$$\rho_{Glycerol} = 1261 \text{ kg/m}^3 [2]$$

$$\mu_{Glycerol} = 1.2 \text{ Pa.s} [3]$$

$$\text{fraction of Glycerol} = 0.065$$

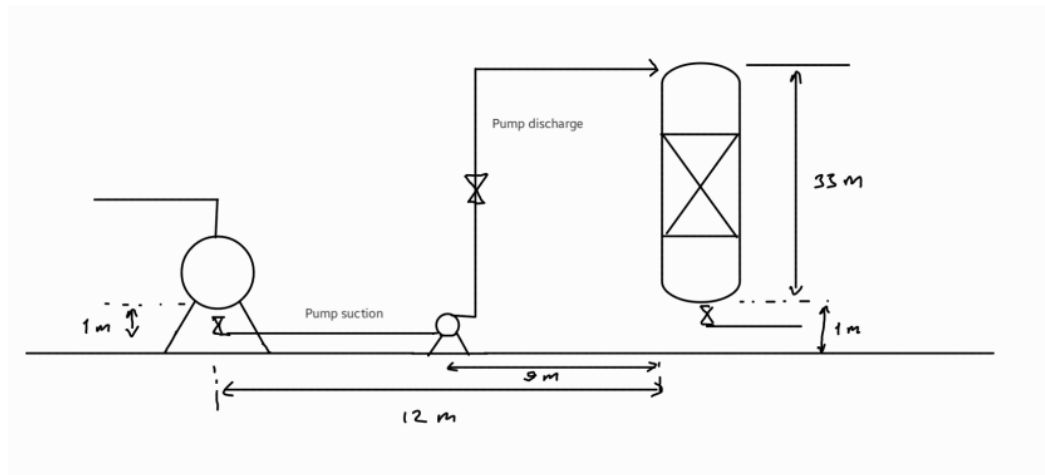
$$\frac{1}{\rho_{mix}} = \sum_i \frac{w_i}{\rho_i}$$

$$\rho_{mix} = \frac{1}{\frac{0.935}{883} + \frac{0.065}{1261}} = 900.546735 \text{ kg/m}^3$$

$$\mu_{mix} = 0.0065 \text{ Pa.s} * 0.935 + 1.2 \text{ Pa.s} * 0.065 = 0.084 \text{ Pa.s}$$

$$\dot{m} = 15000 \text{ kg/h} = 4.167 \text{ kg/s}$$

$$Q = (4.167 \text{ kg/s}) / (900.546735 \text{ kg/m}^3) = 0.0046 \text{ m}^3/\text{s}$$



This process must have a volumetric flow rate of $0.0046 \text{ m}^3/\text{s}$. When designing pipes for such processes, the diameters of the pipes at the pump inlet and outlet are different. To reduce the risk of pump cavitation, the suction line is generally wider than the pump discharge line. A velocity range of 1.5 to 3 m/s is used for pump discharge and 0.9 to 2.4 m/s for pump suction. The maximum and minimum diameters required for these ranges are determined, and then one of the standard pipes is selected from the calculated diameter ranges.

For Pump Section

$$(0.0046\text{m}^3/\text{s})/(0.9\text{m/s}) = 0.0051\text{ m}^2 = a$$

$$\text{inner cross section area} = \pi * d^2/4$$

$$d = \sqrt{4 * a/\pi}$$

$$d = \sqrt{4 * 0.0051\text{m}^2/\pi} = 0.0806\text{ m inner diameter}$$

$$(0.0046\text{m}^3/\text{s})/(2.4\text{m/s}) = 0.00192\text{ m}^2 = a$$

$$d = \sqrt{4 * 0.00192\text{ m}^2/\pi} = 0.050\text{ m inner diameter}$$

The minimum pipe diameter should be 0.050 m and the maximum should be 0.0806 m. According to calculations, a Schedule 40 pipe with a nominal pipe size of 2.5 inches seems appropriate. This pipe has an inner diameter of 62.71 mm and a thickness of 5.16 mm.

For Pump Discharge

$$(0.0046\text{m}^3/\text{s})/(1.5\text{m/s}) = 0.0031\text{ m}^2 = a$$

$$\text{inner cross section area} = \pi * d^2/4$$

$$d = \sqrt{4 * a/\pi}$$

$$d = \sqrt{4 * 0.0031\text{m}^2/\pi} = 0.0628\text{ m inner diameter}$$

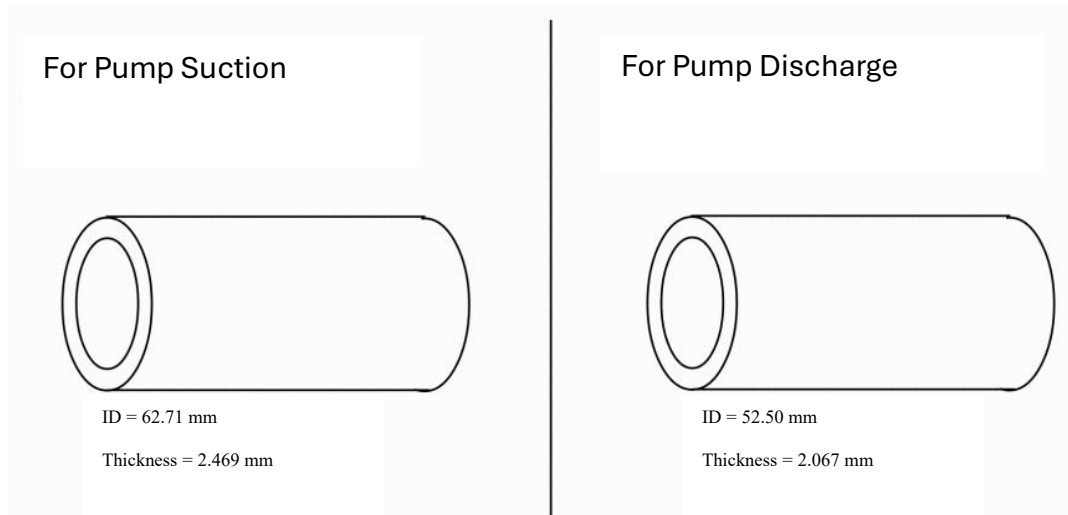
$$(0.0046\text{m}^3/\text{s})/(3\text{ m/s}) = 0.00153\text{ m}^2 = a$$

$$d = \sqrt{4 * 0.00153\text{ m}^2/\pi} = 0.041\text{ m inner diameter}$$

The minimum pipe diameter should be 0.041 m, and the maximum should be 0.0628 m. According to these calculations, Schedule 40 pipe with a 2-inch nominal pipe size can be used. This pipe has an inner diameter of 52.5 mm and a thickness of 3.91 mm. Extra 2 mm corrosion allowance can be added to these thickness values. The pipe material was chosen as stainless steel because it has low friction.

According to ASTM standards, the term Schedule (SCH) indicates the wall thickness classification of a pipe. SCH 40 is one of the most commonly used pipe schedules and is suitable for medium pressure and temperature conditions. Therefore, it provides an optimal balance of safety and cost-effectiveness in process line applications. The selected 2-inch SCH 40 and 2.5-inch SCH 40 pipes were found suitable in terms of:

Mechanical strength, Corrosion resistance, Pressure carrying capacity, Compatibility with standard equipment and fittings. In addition, their wide availability in the market ensures advantages in terms of procurement and maintenance.



$$a = \pi * (0.06271 \text{ m})^2 / 4 = 0.0030886 \text{ m}^2$$

$$Q/a = v$$

$$v_{1(\text{pump suction})} = (0.0046 \text{ m}^3/\text{s}) / (0.0030886 \text{ m}^2) = 1.5 \text{ m/s}$$

$$a = \pi * (0.0525 \text{ m})^2 / 4 = 0.00216 \text{ m}^2$$

$$v_{2(\text{pump discharge})} = (0.0046 \text{ m}^3/\text{s}) / (0.00216 \text{ m}^2) = 2.15 \text{ m/s}$$

K Values

Well-rounded entrance K=0.04

Exit K=1

2 * 90° flanged K=2 * 0.3

Fully open gate valve K=0.15

$$\Sigma K = 1.79$$

$$Re(\text{pump suction}) = \frac{\rho v d}{\mu} = \frac{900.546735 \text{ kg/m}^3 \times 1.5 \text{ m/s} \times (62.71 \times 10^{-3} \text{ m})}{0.084 \text{ Pa.s}} = 1008.451$$

$$Re(pump\ discharge) = \frac{\rho v d}{\mu} = \frac{900.546735 kg/m^3 \cdot 2.15 m/s \times (62.71 \times 10^{-3} m)}{0.084 Pa.s}$$

$$= 1445.447$$

$$1008.451 < 2100 \Rightarrow \text{Laminar}$$

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For pump suction:

$$L = 3m$$

$$F_f = 16/Re = 16/1008.451 = 0.0159$$

$$F_{straight} = 4 \times F_f \times \frac{\mathcal{L} v^2}{2d} = \frac{4(0.0159) \times 3 m (1.5 m/s)^2}{62.71 \times 10^{-3} m \cdot 2} = 1.141 m^2/s^2$$

For pump discharge:

$$L = 42m$$

$$F_f = 16/Re = 16/1445.447 = 0.0111$$

$$F_{straight} = 4 \times F_f \times \frac{\mathcal{L} v^2}{2d} = \frac{4(0.0111) \times 42m (2.15 m/s)^2}{52.5 \times 10^{-3} m \cdot 2} = 82.0956 m^2/s^2$$

$$K = 1.79$$

$$F_{minor} = K \frac{v^2}{2} = \frac{1.79 (2.15)^2}{2} = 4.1371 m^2/s^2$$

$$\Sigma F = 4.1371 + 82.0956 + 1.141 = 87.3737 m^2/s^2$$

The flow inside the tube is in laminar regime; thus, the Fanning friction factor is given by the equation below:

$$f_F = \frac{16}{Re}$$

It is believed that the frictional losses in straight pipe sections can be given using the following equation:

$$\sum F_{straight} = \left(\frac{4f_F L}{d_{tube}} \right) \left(\frac{v^2}{2} \right)$$

And minor losses due to fittings, valves etc.,

$$\sum F_{minor} = \left(\sum K \right) \left(\frac{v^2}{2} \right)$$

$$\sum K = K_{elbows} + K_{valves} + K_{exit} + K_{enterance} + K_{expansion/contraction} + K_{pipe\ exit\ etc.}$$

Steady-state Mechanical Energy Balance for incompressible flow (modified Bernoulli equation)

$$g(Z_2 - Z_1) + \left[\left(\frac{v_2^2}{2} \right) - \left(\frac{v_1^2}{2} \right) \right] + \left[\left(\frac{P_2}{\rho} \right) - \left(\frac{P_1}{\rho} \right) \right] - \widehat{W}_s + \sum F = 0$$

where

$$\sum F = \sum F_{straight} + \sum F_{minor}$$

$$z_2 - z_1 = 33 \text{ m}$$

$$v_1 = 1.5 \text{ m/s}$$

$$v_2 = 2.15 \text{ m/s}$$

$$\Delta P = p_2 - p_1 = 496422.5247 \text{ Pa}$$

$$\rho = 900.54.6735 \text{ kg/m}^3$$

$$Total F_{straight} = 1.141 + 82.0956 = 83.2366 \text{ m}^2/\text{s}^2$$

$$F_{minor} = 4.1371 \text{ m}^2/\text{s}^2$$

$$Total F = 87.3737 \text{ m}^2/\text{s}^2$$

Mechanical Energy Balance

$$\begin{aligned} & 9.81 \text{ m/s}^2 * (33 \text{ m}) + [(2.15 \text{ m/s})^2 - (1.5 \text{ m/s})^2]/2 \\ & + [496422.5247 \text{ pa} / 900.546735 \text{ kg/m}^3] - W_s \\ & + 87.3737 \text{ m}^2/\text{s}^2 = 0 \end{aligned}$$

$$323.73 \text{ m}^2/\text{s}^2 + 2.3725 \text{ m}^2/\text{s}^2 + 551.2457 \text{ m}^2/\text{s}^2 - W_s + 87.3737 \text{ m}^2/\text{s}^2 = 0$$

$$\text{m}^2/\text{s}^2 [=] \text{ joule/kg}$$

$$W_s = 964.7219 \text{ joule/kg}$$

Assume with %75 efficiency of pump;

$$w_s = w_p * efficiency$$

$$w_p = 964.7219/0.75 = 1286.3 \text{ joule/kg}$$

$$1286.3 \text{ Joule/kg} * 15000 \text{ kg/h} = 19294500 \text{ joule/h} = 5.36 \text{ kW}$$

The electricity price for businesses is TRY 4.350 / kWh [4]

$$5.36 \text{ kW} * 4.350 \text{ TRY} = 23.316 \text{ TRY/hour} = 16787.5 \text{ TRY/month}$$

Properties	Value	Unit
Density of Biodiesel	883	kg/m ³
Density of Glycerol	1261	kg/m ³
Viscosity of Biodiesel	0.0065	Pa.s
Viscosity of Glycerol	1.2	Pa.s
Density of Mix	900.546735	kg/m ³
Viscosity of Mix	0.084	Pa.s
Volumetric Flow Rate	0.0046	m ³ /s

Calculation Results	Value	Unit
Pump Suction Pipe	2.5-inch SCH 40	
Pump Discharge Pipe	2-inch SCH 40	
Pipe Material	Stainless Steel	
ID of pump suction	62.71	mm
ID of pump Discharge	52.5	mm
Thickness of pump suction	5.16	mm
Thickness of pump Discharge	3.91	mm
Velocity of pump suction	1.5	m/s
Velocity of pump Discharge	2.15	m/s
Total K value	1.79	
Pump Suction Reynold's Number	1008.451	
Pump Discharge Reynolds Number	1445.447	
Total F_{Straight}	83.2366	m^2/s^2
F_{minor}	4.1371	$\text{m}^2/\text{s}^{131}$
Total Friction Loss	87.3737	$\text{m}^2/\text{s}^{132}$
Shaft Work	964.7219	joule/kg
Efficiency	75%	
Pump Work	1286.3	joule/kg
Electric Cost for Pump	16787.5	TRY/month

Environmental and Safety Considerations

When designing the piping system for biodiesel–glycerol mixtures, both environmental and safety aspects must be considered. From an environmental standpoint, biodiesel is biodegradable and less toxic than conventional diesel, which reduces risks in the event of accidental leaks. However, secondary containment systems and the use of corrosion-resistant stainless-steel pipes are still necessary to prevent soil and water contamination. In addition, energy-efficient pump operation helps lower overall electricity consumption, indirectly reducing environmental impact. From a safety perspective, biodiesel and its blends may present flammability hazards depending on their flash point, especially when mixed with diesel or kerosene. Therefore, proper labeling, fire prevention measures, and spill control strategies should be implemented. Maintaining adequate ventilation, providing safety signage, and integrating pressure relief valves are also essential to avoid vapor accumulation and excessive pressure buildup.[5]

Conclusion

In conclusion, the fluid system was successfully designed by considering both hydraulic and economic aspects. The selection of 2.5-inch SCH 40 pipe for suction and 2-inch SCH 40 pipe for discharge, in accordance with ASTM standards, ensured that flow velocities and frictional losses remained within acceptable limits. The use of stainless steel further enhanced system durability and efficiency. The overall design achieved safe operation, minimized cavitation risk, and optimized pumping costs, thereby fulfilling the project objectives.

References

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